

ECE 370 Electromagnetic Fields and Waves

HOMEWORK #1

I expect you to clearly show me how you solved these problems – an orderly and visible presentation of your thoughts and process toward solution. I want to easily see that you understand what you are doing.

Name: Gianni Avilan-Losee

1- Find the results of the following products of unit vectors. Show your work clearly.

a) $\hat{a}_\phi \cdot \hat{a}_x$

b) $\hat{a}_R \cdot \hat{a}_y$

c) $\hat{a}_r \times \hat{a}_R$

d) $\hat{a}_\theta \times \hat{a}_z$

2- Vectors $\vec{A}$ and $\vec{B}$ are given in rectangular coordinates by

$$\vec{A} = 4\hat{x} + 2\hat{y} - \hat{z}$$

$$\vec{B} = -2\hat{x} + 3\hat{z}$$

Find:

a) The scalar component, or projection, of $\vec{B}$ in the direction of $\vec{A}$.

b) The vector component of $\vec{B}$ in the direction of $\vec{A}$.

c) The vector component of $\vec{B}$ perpendicular to $\vec{A}$.

3- Transform the rectangular coordinate point $P = (1,2,3)$ into cylindrical and spherical coordinates.

Problem	Course outcome	Grade
1	1-d	/40
2	1-d	/40
3	1-d	/20
TOTAL:		/100

1. a) In the cylindrical coordinate system, $\hat{\alpha}_r$'s direction is a function of the angle ϕ , as measured counterclockwise from the positive x-axis, and restricted to the range $0 \leq \phi < 2\pi$. In summary, we may assume an angle ϕ between $\hat{\alpha}_x$ and $\hat{\alpha}_r$.

Next, we may assume an angle of 90° between $\hat{\alpha}_r$ and $\hat{\alpha}_\phi$, since all basis vectors in the cylindrical coordinate system are orthogonal, and so define the angle between $\hat{\alpha}_x$ and $\hat{\alpha}_\phi$ as $\phi + 90^\circ$. By the definition of the dot product, $\hat{\alpha}_\phi \cdot \hat{\alpha}_x = |\hat{\alpha}_\phi||\hat{\alpha}_x|\cos(\theta_{\hat{\alpha}_\phi\hat{\alpha}_x})$, where $\theta_{\hat{\alpha}_\phi\hat{\alpha}_x}$ is the angle between the two vectors. Since the magnitude of both vectors $|\hat{\alpha}_r| = |\hat{\alpha}_\phi| = 1$,

$$\hat{\alpha}_\phi \cdot \hat{\alpha}_r = \cos(\theta_{\hat{\alpha}_\phi\hat{\alpha}_r}) = \cos(\phi + 90^\circ) = -\sin(\phi).$$

We can double check this result by using the coordinate system transforms given in Table 3-2 of the textbook, where $\hat{\alpha}_x = \hat{\alpha}_r\cos(\phi) - \hat{\alpha}_\phi\sin(\phi)$; and so, keeping in mind that the dot product is distributive,

$$\hat{\alpha}_\phi \cdot \hat{\alpha}_x = \hat{\alpha}_\phi \cdot (\hat{\alpha}_r\cos(\phi) - \hat{\alpha}_\phi\sin(\phi)) = \hat{\alpha}_\phi \cdot \hat{\alpha}_r\cos(\phi) - \hat{\alpha}_\phi \cdot \hat{\alpha}_\phi\sin(\phi).$$

Note that multiplying either unit vector by a scalar does not affect the angular relationship between it and another vector. Since $\hat{\alpha}_\phi$ and $\hat{\alpha}_r$ are orthogonal, their dot product is equal to 0, while the dot product between the identical vectors $\hat{\alpha}_\phi$ is equal to 1. Thus, the dot product simplifies to

$$\hat{\alpha}_\phi \cdot \hat{\alpha}_r = -\sin(\phi).$$

- b) $\hat{\alpha}_R$'s direction is a function of both the zenith angle θ and the azimuth angle ϕ , similar to how $\hat{\alpha}_r$ is determined relative to ϕ in the cylindrical coordinate system. In this case, we can locate the unit vector $\hat{\alpha}_R$ at an angle θ from the positive z-axis and an angle ϕ from the positive x-axis in the rectangular system.

Transforming $\hat{\alpha}_R$ into the rectangular coordinate system,

$$\hat{\alpha}_R = \hat{\alpha}_x\sin(\theta)\cos(\phi) + \hat{\alpha}_y\sin(\theta)\sin(\phi) + \hat{\alpha}_z\cos(\theta).$$

Once again applying the distributive property of the dot product and reducing orthogonal vectors to zero, as in question 1a, we are left with

$$\hat{\alpha}_R \cdot \hat{\alpha}_y = \sin(\theta)\sin(\phi).$$

- c) First, defining any cross product $\vec{A} \times \vec{B} = |A||B|\sin(\theta_{AB})\hat{n}$, where $\hat{n}$ is the unit normal vector, whose direction is found using the right hand rule, we assume as before that, since $\hat{\alpha}_r$ and $\hat{\alpha}_R$ are unit vectors and have a magnitude of 1, their cross product simplifies to $\hat{\alpha}_r \times \hat{\alpha}_R = \sin(\theta_{AB})\hat{n}$.

The cross product $\vec{A} \times \vec{B}$ may also be expressed in the form of a determinant (in cylindrical coordinates), as

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\alpha}_r & \hat{\alpha}_\phi & \hat{\alpha}_z \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$$

Transforming the unit vector of $\hat{\alpha}_R = \hat{\alpha}_r \sin(\theta) + \hat{\alpha}_z \cos(\theta)$, we may express the cross product between the two given unit vectors as

$$\hat{\alpha}_r \times \hat{\alpha}_r = \hat{\alpha}_r \times \hat{\alpha}_r \sin(\theta) + \hat{\alpha}_r \times \hat{\alpha}_z \cos(\theta).$$

Assuming that the cross product between the identical vectors $\hat{\alpha}_r$ equals zero, irrespective of either vector's scaling, we may then express the cross product in determinant form as

$$\hat{\alpha}_r \times \hat{\alpha}_z \cos(\theta) = \begin{vmatrix} \hat{\alpha}_r & \hat{\alpha}_\phi & \hat{\alpha}_z \\ A_r & A_\phi & A_z \\ B_r \cos(\theta) & B_\phi \cos(\theta) & B_z \cos(\theta) \end{vmatrix}.$$

Since both quantities are unit vectors, the entire determinant simplifies to $-\hat{\alpha}_\phi \cos(\theta)$. Therefore,

$$\hat{\alpha}_r \times \hat{\alpha}_R = -\hat{\alpha}_\phi \cos(\theta).$$

This concurs with the use of the right-hand rule, which indicates that any cross-product between the two unit vectors should be orthogonal to the ϕ - z plane, depending on the order in which the cross product is taken.

- d) In this case, we may transform $\hat{\alpha}_\theta$ into its cylindrical components as $\hat{\alpha}_\theta = \hat{\alpha}_r \cos(\theta) - \hat{\alpha}_z \sin(\theta)$, which expands the given cross product to

$$\hat{\alpha}_r \cos\theta - \hat{\alpha}_z \sin(\theta) \times \hat{\alpha}_z.$$

Making use of the anti-commutative and distributive properties of the cross product, this may be further simplified to

$$-[\hat{\alpha}_z \times \hat{\alpha}_r \cos(\theta) + \hat{\alpha}_z \times \hat{\alpha}_z \sin(\theta)] = \hat{\alpha}_z \times \hat{\alpha}_r \cos(\theta).$$

Using the determinant definition of the dot product as before,

$$\hat{\alpha}_z \times \hat{\alpha}_z \cos(\theta) = \begin{vmatrix} \hat{\alpha}_r & \hat{\alpha}_\phi & \hat{\alpha}_z \\ 0 & 0 & -1 \\ \cos(\theta) & 0 & 0 \end{vmatrix} = -\hat{\alpha}_\phi \cos(\theta),$$

which also corresponds with the directional component given by the right-hand rule.

2. a) The scalar component of $\vec{B}$ in the direction of $\vec{A}$ is found by means of the dot product between $\vec{B}$ and the unit vector pointing in the direction of $\vec{A}$. First, finding the unit vector pointing in the direction of $\vec{A}$,

$$\hat{\alpha}_A = \frac{\vec{A}}{|\vec{A}|} = \hat{\alpha}_x \frac{4}{\sqrt{21}} + \hat{\alpha}_y \frac{2}{\sqrt{21}} - \hat{\alpha}_z \frac{1}{\sqrt{21}}.$$

Then, the scalar component of $\vec{B}$ in the direction of $\vec{A}$ is

$$\vec{B} \cdot \hat{\alpha}_A = \frac{-8}{\sqrt{21}} - \frac{3}{\sqrt{21}} = -\frac{11}{\sqrt{21}}.$$

- b) To find the vector component or projection of $\vec{B}$ in the direction of $\vec{A}$, we must find the vector whose magnitude is the scalar projection of $\vec{B}$ on $\vec{A}$ with the same direction of $\vec{A}$. To find this, we may simply scale the unit vector pointing in the $\vec{A}$ direction by the scalar component of $\vec{B}$ in the $\vec{A}$ direction,

$$(\vec{B} \cdot \hat{\alpha}_A) \hat{\alpha}_A = -\hat{\alpha}_x \frac{44}{21} - \hat{\alpha}_y \frac{22}{21} + \hat{\alpha}_z \frac{11}{21}.$$

- c) Finally, to find the vector component of $\vec{B}$ in the direction of A , we may use the property that the sum of $\vec{B}$'s vector component parallel to and perpendicular to the direction of A will equal $\vec{B}$. In other words, $\vec{B}$ is made up of a component parallel to the $\vec{A}$ direction and another component orthogonal to the $\vec{A}$ direction. Therefore, the component orthogonal to the $\vec{A}$ direction is given by

$$\vec{B} - [(\vec{B} \cdot \hat{\alpha}_A) \hat{\alpha}_A] = \hat{\alpha}_x \frac{2}{21} + \hat{\alpha}_y \frac{22}{21} + \hat{\alpha}_z \frac{52}{21}.$$

3. Given the point $P(1, 2, 3)$ in rectangular coordinates, we may first convert P to cylindrical coordinates of the form $P(r, \phi, z)$ by using the coordinate transformations $r = \sqrt{x^2 + y^2}$, $\phi = \text{atan2}(\frac{y}{x})$, and $z = z$. Therefore,

$$P_{\text{rectangular}}(1, 2, 3) = P_{\text{cylindrical}}(2.236, 63.43^\circ, 3).$$

Converting to spherical coordinates, we may use the coordinate transformations $R = \sqrt{x^2 + y^2 + z^2}$, $\theta = \text{atan2}(\frac{\sqrt{x^2 + y^2}}{z})$, and $\phi = \text{atan2}(\frac{y}{x})$. Therefore,

$$P_{\text{rectangular}}(1, 2, 3) = P_{\text{cylindrical}}(3.7417, 36.70^\circ, 63.43^\circ).$$